

DATA

inputs and references

TBannotator

≈274 800 MTBC genomes

L4.13 cohort

n = 258 strains
(61 + 197 sub-clades)

Sister L4 panel

n = 1 485 strains
(17 sub-lineages)

H37Rv reference

NC_000962.3
GenBank + GFF3

STRAIN-LEVEL ANALYSES

per-genome QC, classification, metadata

Quality control

depth, coverage, GC,
contamination

Lineage classification

17 published schemes
(Coll, Stucki, Napier,
Freschi, ...)

In silico spoligotyping

43 spacers,
SITVIT2 lookup

ENA metadata

country, host,
collection date,
BioProject

LINEAGE-LEVEL ANALYSES

phylogeny, selection, geography, dynamics

Phylogeny

RAxML-NG, BIN+G,
100 bootstraps

Core-exclusive SPDI

≥95 % in L4.13,
absent in sisters

MK + convergence

6 strata, DoS,
permutation null

Ancestral states

ML marginal (MPPA),
8 regions

Molecular dating

IQ-TREE + LSD2,
synonymous sites

Time-scaled HD

Rasigade 2017,
 $t_{50} = 20 / 200$ yr

OUTPUTS

markers, figures, tables, supplementary

3 SNP markers

L4.13 / .1 / .2
(Table 1)

Phylogenetic tree

Figure 2

Geographic maps

Figure 3

THD scores

Figure 4

MK results

Table 2 + Supp.